

VerbalVis: Full-Duplex Conversational Visual Analytics for Analytical Intent Revision

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VerbalVis: Supporting Analytical Intent Evolution through Full-Duplex Conversational Visual Analytics

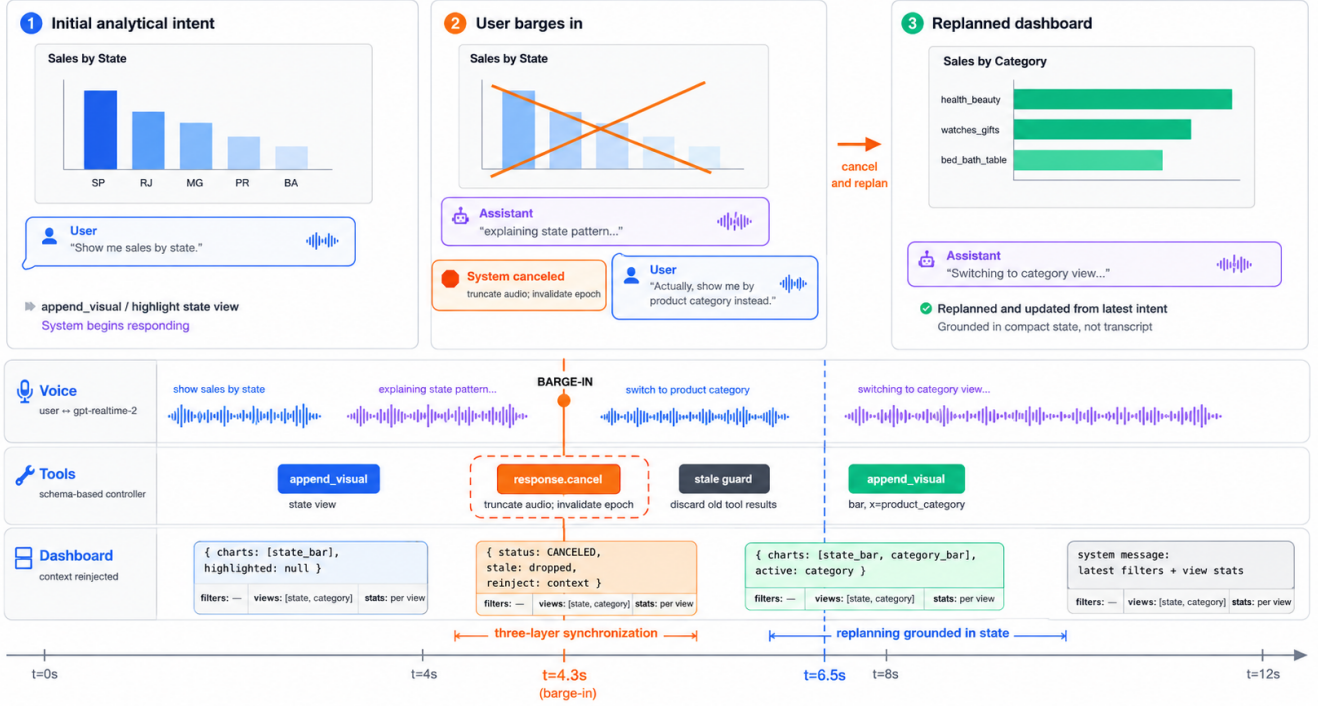


Fig. 1. Overview of a barge-in episode in VerbalVis. While the assistant is explaining a visualization, the user interrupts to redirect the exploratory analysis from sales by state to sales by product category. VerbalVis immediately cancels obsolete speech, replans visualization actions through tool execution, reinjects the updated dashboard state as contextual grounding for the language model, and synchronously updates both the dashboard and spoken response. The aligned voice, tool, and dashboard timelines illustrate the coordinated full-duplex interaction that enables continuous analytical intent evolution without explicit turn switching.

Abstract—Exploratory data analysis is iterative: new observations may lead analysts to redirect questions, revise provisional explanations, or refine the data scope under investigation. Existing speech- and language-driven visual analytics systems generally support such changes only at discrete turn boundaries. Full-duplex interaction enables users to redirect an analysis while the assistant is still speaking, but stopping audio alone is insufficient when the superseded response may also have initiated actions that modify a persistent dashboard. We present VerbalVis, a full-duplex conversational visual analytics system that coordinates spoken output, schema-grounded tool execution, and dashboard state during mid-response redirection. Drawing on prior work in sensemaking and exploratory analysis, we use three non-exhaustive and potentially overlapping constructs—*Analytical Goal Shift*, *Working-Hypothesis Revision*, and *Analytical Scope Refinement*—to characterize revision episodes. A lightweight formative study refined their operational boundaries and informed requirements for cancellation, state preservation, and visible feedback. VerbalVis truncates obsolete speech, in-

validates superseded tool actions, discards stale results, and replans against the latest committed dashboard state. We evaluate it through an illustrative scenario and an exploratory within-subjects comparison of full-duplex and turn-based interaction.

Index Terms—Visual analytics, conversational interfaces, speech interaction, full-duplex dialogue, analytical intent, exploratory data analysis, large language models.

I. INTRODUCTION

EXPLORATORY data analysis (EDA) is an iterative process in which analysts move between observations, questions, provisional explanations, and new analytical actions. Rather than executing a fully specified plan, analysts often begin with incomplete information needs and revise their direction as patterns, anomalies, and competing explanations emerge [?], [?], [?], [?]. Such revisions may concern different aspects of the investigation. An analyst may replace the

primary question being pursued, reconsider a working explanation, or restrict the analysis to a more relevant population, period, region, or set of variables. We use *analytical intent revision* to refer to a local episode in which the analyst supersedes, qualifies, or materially modifies an active analytical commitment. Across an exploratory session, the accumulation of such episodes constitutes the broader evolution of analytical intent.

Natural-language interfaces for visualization have made it increasingly possible to express analytical operations through conversational commands. Systems such as DataTone, Eviza, and NL4DV map language to data attributes, filters, comparisons, aggregations, and visual specifications, while multimodal systems combine language with pointing, touch, or direct manipulation [?], [?], [?], [?], [?]. More recent LLM-powered visual analytics systems extend this interaction model to data transformation, multistep planning, visualization generation, tool execution, and conversational refinement [?], [?], [?], [?]. These systems substantially increase the range of analytical requests that can be expressed in language. However, interaction generally remains organized around discrete handoff points: the user submits a request, the system produces a response or completes a sequence of actions, and the user subsequently provides a follow-up or correction.

This turn-based organization is particularly consequential for spoken interaction because system output unfolds over time. While listening to an assistant’s explanation, an analyst may recognize that the system is pursuing an irrelevant question, disagree with a tentative explanation, or identify a more appropriate data subset. Requiring the analyst to wait until the current response has finished separates the moment at which the analytical direction changes from the moment at which the system can act on that change. Recent full-duplex speech models provide a different interaction regime: the system can continue listening while speaking, allowing the user to reclaim the floor through barge-in. Yet barge-in alone does not solve the visual analytics problem. Barge-in is a temporal event—the user begins speaking during system output—whereas analytical revision is a semantic relation between the new utterance and the currently active investigation. A user may interrupt only to repair a recognition error, request clarification, or control the conversation; conversely, a user may revise an analysis after waiting for the assistant to finish.

More importantly, stopping audio does not necessarily stop the analytical consequences of the superseded response. A conversational visual analytics system may already have generated a tool call, initiated a database operation, created a visualization specification, or scheduled a dashboard update. These actions can complete after the user’s latest utterance has redirected the analysis. If their results are committed without checking whether the originating response remains valid, the dashboard may continue to evolve according to an obsolete intent even after the assistant has verbally acknowledged the new request. Full-duplex conversational visual analytics therefore introduces a state-consistency problem spanning three coupled layers: spoken output, analytical tool execution, and persistent visualization state. Supporting interruption at the audio layer is insufficient unless response-dependent analytical work can

also be invalidated and the revised request can be interpreted against the latest valid dashboard state.

We present **VerbalVis**, a full-duplex conversational visual analytics system designed to support analytical redirection while system speech and analytical actions are still unfolding. VerbalVis integrates a real-time speech model with schema-grounded visualization tools and a persistent dashboard. When new user speech begins during an active response, the system immediately stops local audio playback and requests cancellation of the obsolete response. Each response and its associated tool actions are bound to an execution epoch. A redirection advances the current epoch, causing tool results associated with superseded responses to be rejected rather than committed to the dashboard. After the redirecting utterance is interpreted, VerbalVis replans from the latest committed dashboard state, including active filters, highlights, and visualizations. A compact representation of this state is reinjected as conversational context so that subsequent speech and tool execution remain grounded in what the analyst currently sees.

VerbalVis does not introduce a new speech model or interruption detector. Its contribution lies in the visual-analytics-specific orchestration required to make full-duplex redirection coherent across speech, computation, and visual state. In particular, the system treats the latest valid request as potentially superseding an unfolding response and coordinates speech truncation, response cancellation, stale-action invalidation, late-result rejection, and dashboard-state-grounded replanning. This mechanism is agnostic to the particular reason for the redirection: VerbalVis does not classify revision types at runtime, and all superseded response work is governed by the same response-lifecycle and state-validity safeguards.

To inform the design and analysis of this interaction, we draw on prior work in sensemaking, scientific reasoning, information seeking, and exploratory visual analysis. These traditions independently describe evolving analytical goals, revisable hypotheses or explanatory frames, and progressive focusing of broad information needs [?], [?], [?], [?], [?], [?]. We synthesize these traditions into three theory-informed objects of analytical revision: *Analytical Goal Shift*, in which the primary analytical question or desired knowledge outcome is materially reoriented; *Working-Hypothesis Revision*, in which a provisional explanation is rejected, replaced, or qualified; and *Analytical Scope Refinement*, in which constraints over the relevant population, time period, geography, variables, categories, granularity, or data subset are changed while the broader objective is generally preserved. These constructs are non-exhaustive and potentially overlapping. They provide a vocabulary for formative inquiry, behavioral coding, and design discussion, rather than a universal taxonomy of EDA behavior or a set of runtime intent classes.

We conducted a lightweight formative design study with four participants experienced in data analysis. The study did not attempt to derive or validate a comprehensive taxonomy. Instead, it used the three constructs as theory-informed sensitizing concepts to examine how analytical redirection is expressed in speech, which aspects of prior context users expect to preserve, which obsolete actions they expect the system to cancel, and which dashboard operations are needed to

enact a revised direction. The sessions also exposed important boundary cases, including recognition repair, ordinary follow-up, requests for additional evidence, and changes to analytical method or visual representation. These observations refined the operational definitions used in our analysis and informed requirements for semantic interpretation, cross-layer cancellation, state-grounded replanning, visible execution state, and recoverable dashboard manipulation.

We evaluate VerbalVis through an illustrative usage scenario and an exploratory within-subjects mixed-methods study with eight participants. The study compares a *Full-Duplex* condition, in which users may speak while the assistant is responding, with an otherwise matched *Turn-Based* condition, in which users must wait for the current response to finish. Both conditions use the same model, voice, dashboard, analytical tools, tasks, and initial state. We examine system responsiveness and state consistency, perceived control, interaction flow, workload, exploratory behavior, analytical outcomes, and the ways in which participants naturally revise an ongoing analysis. The three analytical revision constructs are used to characterize observed episodes, but the study is designed to evaluate the interaction and orchestration mechanism rather than to validate the constructs as a universal classification.

Contributions.

- 1) A theory-informed characterization of analytical revision in conversational visual analysis, distinguishing changes to the analytical goal, working hypothesis, and analytical scope, together with formative design evidence and requirements for supporting mid-response redirection.
- 2) VerbalVis, a full-duplex conversational visual analytics system that coordinates speech truncation, response supersession, schema-grounded tool execution, stale-result invalidation, and replanning against the latest committed dashboard state.
- 3) An exploratory mixed-methods evaluation comparing full-duplex and turn-based speech interaction in open-ended visual analysis, examining interaction responsiveness, perceived control, workload, exploratory behavior, and analytical outcomes.

II. RELATED WORK

VerbalVis lies at the intersection of natural-language interaction for visualization, LLM-powered visual analytics, full-duplex spoken dialogue, and research on sensemaking and analytical revision. Prior work in these areas provides important foundations for language grounding, multimodal interaction, agent planning, interruption handling, and the representation of evolving analysis. However, these capabilities have largely been studied separately. VerbalVis focuses on their coordination when a spoken request semantically supersedes an active response whose speech, analytical actions, and dashboard updates may still be unfolding.

A. Natural-Language Interaction for Visualization

Visualization-oriented natural-language interfaces (V-NLIs) seek to translate user utterances into references to data attributes, values, analytical tasks, and visual encodings. Early

systems established that language could serve as more than a textual shortcut to chart construction. DataTone, for example, represents alternative interpretations through interactive ambiguity widgets, allowing users to inspect and resolve underspecified mappings rather than silently accepting a single parse [?]. Eviza grounds utterances in both the underlying dataset and the active visualization, supporting references to visible marks, filters, aggregations, and contextual follow-up questions [?]. NL4DV generalizes this translation process into a reusable toolkit that extracts data attributes and analytical tasks and produces candidate visualization specifications [?]. Together, these systems establish language grounding, ambiguity management, and contextual interpretation as central requirements for conversational visual analysis.

Subsequent work expands language interaction through multimodal input and richer conversational context. Orko combines speech and direct manipulation for network exploration, enabling users to express analytical operations through complementary modalities [?]. InChorus develops a systematic vocabulary of operations, targets, parameters, and instruments for interactions combining speech, pen, and touch across visualization types [?]. DataBreeze similarly interweaves spoken language with direct manipulation over flexible unit visualizations, allowing users to alternate between systematically encoded views and manually arranged visual structures [?]. Snowy and conversational extensions to NL4DV further investigate follow-up utterances, context-dependent references, and the modification of existing views [?], [?]. These systems demonstrate that speech can complement visual interaction and that later utterances can be interpreted relative to earlier dialogue and visualization state.

Other V-NLIs address increasingly complex analytical questions. FlowSense integrates natural language into a dataflow environment, grounding requests in the structure and state of an analytical workflow. Talk2Data decomposes complex questions into simpler analytical requests and presents their answers through multiple annotated visualizations [?]. Such systems move beyond one-shot chart generation toward contextual and multi-step analysis. Nevertheless, conversational refinement is generally organized around completed input and output units: an utterance is interpreted, a visualization or response is produced, and the user subsequently refines the result. Multimodal input may reduce the effort required to express an operation, but speech is usually treated as an alternative or complementary command channel rather than as a continuously available channel through which an unfolding response can be superseded.

This distinction matters because modifying an analysis after a response has completed is different from redirecting it while speech and computation are still active. Existing V-NLIs provide important mechanisms for language grounding, ambiguity resolution, follow-up interpretation, and multimodal reference. They do not, however, generally specify how a new spoken request should invalidate response-dependent actions that have already been generated but have not yet been committed to a persistent dashboard. VerbalVis builds on their contextual grounding and schema-based execution, but shifts attention from turn-level refinement to the lifecycle of an active

analytical response.

B. LLM-Powered and Agentic Visual Analytics

Large language models expand conversational data analysis from mapping utterances to individual visual specifications toward data transformation, code generation, multistep planning, tool execution, and insight interpretation. LIDA decomposes visualization generation into stages for data summarization, goal generation, visualization creation, and evaluation [?]. Data Formulator combines natural-language concepts with directly specified visual encodings, allowing an LLM to synthesize the data transformations required to instantiate a user-defined visualization [?]. Data Formulator 2 extends this approach to iterative visualization authoring by supporting blended natural-language and interface input, the reuse of prior designs, and navigation through a branching history of data and chart variants [?]. DynaVis similarly augments generated visualizations with persistent editing controls, supporting continued refinement beyond the initial prompt [?].

Agentic visual analytics systems further delegate analytical planning and execution to LLM-based components. InsightPilot and related systems use language models to recommend or conduct sequences of data-analysis operations rather than generating only a single chart [?]. LightVA organizes analysis through planner, executor, and controller components that translate high-level analytical goals into lower-level tasks, visualizations, and insights [?]. Its task-flow representation also gives users a means to inspect and manage the developing analytical plan. ProactiveVA extends agent assistance from reactive question answering toward context-sensitive intervention based on the user’s ongoing interaction history [?]. These systems demonstrate how LLM agents can participate in the planning and execution of visual analysis rather than merely serving as language parsers.

A related line of work improves the inspectability and steerability of LLM-mediated analysis. WaitGPT transforms generated analysis code into an incrementally presented visual representation through which users can inspect, verify, and modify individual operations while an analysis is being produced [?]. InsightLens records, organizes, and links insights, visualizations, code, and conversational context to support navigation through an extended LLM-assisted analysis [?]. Sensecape and other nonlinear conversational interfaces allow users to branch, spatially organize, and revisit conversational material rather than preserving a single linear transcript [?]. These systems respond to limitations of monolithic chat interfaces by externalizing intermediate reasoning, analytical history, and alternative branches.

This work establishes several capabilities required by VerbalVis: LLM-generated analytical plans, schema- or code-based execution, persistent artifacts, history management, and user steering. However, steering is typically expressed through text, graphical controls, or intervention at an inspectable checkpoint. Even when intermediate operations are streamed, the interaction model need not address concurrent spoken overlap or the possibility that a late result from a superseded response will modify the current dashboard. Branching histories preserve alternatives, while provenance views make prior

steps inspectable, but neither capability by itself defines which in-flight actions remain authorized to commit after the user has changed direction.

VerbalVis therefore addresses a complementary orchestration problem. It does not seek to improve general-purpose LLM planning or autonomous visualization generation. Instead, it binds generated analytical actions to the lifecycle of the response that produced them. When that response is superseded, associated actions become stale, and their results are prevented from modifying the dashboard. This connects conversational steering to explicit state-validity semantics across speech, tools, and visual output.

C. Full-Duplex Conversational AI and Interruption

Spoken interfaces vary in how they coordinate listening and speaking. Push-to-talk and conventional turn-based systems separate user and system audio into discrete turns. Streaming systems may incrementally recognize or generate speech, but streaming alone does not imply that the system listens and speaks concurrently. In a full-duplex system, user and assistant audio can overlap: the system continues processing incoming speech while producing its own output and must decide whether the input is background noise, a backchannel, conversational overlap, or an attempt to take the floor [?].

Barge-in has long been studied in spoken-dialogue systems as the ability to detect user speech during system output and stop or modify the current response [?]. Incremental dialogue research further shows that partial recognition hypotheses and dialogue interpretations may be revised as additional speech arrives [?]. More recent full-duplex speech systems model concurrent input and output streams, natural overlap, backchannels, echo cancellation, and context-sensitive interruption [?], [?], [?], [?], [?]. These systems substantially improve the temporal coordination of machine conversation by reducing dependence on rigid end-of-turn detection.

HCI research also demonstrates that interruption policy shapes conversational experience. Systems that ignore legitimate barge-in can make users feel constrained by machine-controlled turns, whereas overly aggressive interruption may terminate useful output in response to acknowledgements, noise, or hesitation. Studies of turn-taking and interruption therefore consider timing, floor management, responsiveness, and perceived control [?], [?]. Conversation analysis provides an additional distinction between interruption and repair: repair practices address trouble in speaking, hearing, or understanding, and do not necessarily change the underlying task or informational goal [?].

These distinctions are essential for conversational visual analytics. Barge-in is an observable timing event, while analytical revision concerns the semantic relationship between a new utterance and the active investigation. A user may barge in to acknowledge the assistant, correct a recognition error, request repetition, or ask the assistant to stop speaking without changing the analysis. Conversely, the user may wait until a response ends and then revise the analytical goal, explanation, or scope. Interruption timing and analytical revision should therefore be represented and evaluated separately.

Existing full-duplex research primarily addresses the speech layer: whether the system continues listening, when it yields the floor, and how generated audio is truncated or resumed. In visual analytics, however, a spoken response may be coupled to database queries, visualization generation, filtering, or other persistent state changes. Audio cancellation does not imply computational cancellation, and computational cancellation does not guarantee that a result already in transit will not be rendered. VerbalVis extends full-duplex interaction from conversational floor management to analytical state management. Speech onset can conservatively suspend obsolete output, while response identifiers and execution epochs determine whether associated tool results remain valid when they later reach the dashboard.

D. Sensemaking and Analytical Revision

Sensemaking and exploratory-analysis research provides the theoretical foundation for treating analytical direction as revisable. Russell et al. describe sensemaking as the construction and adaptation of representations that support task-specific questions, emphasizing the costs of searching for, instantiating, and shifting among representations [?]. Pirolli and Card model intelligence analysis as iterative movement between information foraging and sensemaking activities, including the collection of evidence, construction of schemas, generation of hypotheses, and development of a final representation [?]. In visual analytics, Sacha et al. similarly describe knowledge generation through interacting exploration, verification, and reasoning loops in which findings may generate new questions or support, qualify, and falsify hypotheses [?]. These models do not treat analysis as the execution of a fixed initial plan.

Prior work offers distinct foundations for changes to analytical goals, working explanations, and scope. Empirical and theoretical accounts of information seeking show that initial needs are often incomplete and evolve as information is encountered. Bates' berrypicking model describes search as a sequence of evolving queries rather than the execution of one stable formulation [?]. Vakkari shows how broad information needs become increasingly focused as a research problem develops [?]. In exploratory visual analysis, Battle and Heer synthesize evidence that analysts begin with goals of varying specificity, refine them through interaction, and decompose open-ended analysis into more focused subtasks [?]. These accounts support the possibility of an *Analytical Goal Shift*, while also showing that a complete replacement of the primary question must be distinguished from incremental goal refinement or a change in strategy.

Hypothesis and frame revision provide a second foundation. Data-Frame Theory describes sensemaking as a reciprocal process in which data are interpreted through an explanatory frame and the frame is elaborated, questioned, compared, or replaced as new evidence is encountered [?]. Scientific-reasoning research distinguishes search within a hypothesis space from search for experiments or evidence used to evaluate those hypotheses [?]. This distinction separates a change in explanation from a request for another visualization or test. Research on anomalous evidence further shows that conflicting

observations can lead to theory revision, but may also be ignored, reinterpreted, or judged insufficient [?]. We therefore use *Working-Hypothesis Revision*, rather than hypothesis correction, to avoid implying that the newly proposed explanation is necessarily true.

A third foundation concerns progressive focusing and changes to the data target. Information-foraging models describe the selection of increasingly promising information patches, while search and visualization systems support the progressive specification of facets, populations, variables, and constraints [?], [?], [?]. Brehmer and Munzner's distinction among the analytical *why*, interaction *how*, and data *what* helps separate a change in analytical purpose from a change in the data to which that purpose applies [?]. We use *Analytical Scope Refinement* for changes to population, time, geography, variables, categories, granularity, or data subset that generally preserve the higher-level objective. Filtering is one possible implementation of such a revision, but a filter operation is not itself sufficient evidence that the analyst has semantically revised the scope.

Analytical provenance and history systems capture how such reasoning unfolds. Provenance frameworks distinguish histories of data, visualization state, interaction, and reasoning and support review, communication, recovery, and reproducibility [?]. SensePath combines interaction logs with think-aloud data to reconstruct sensemaking processes [?], while graphical histories and branching systems support revisiting prior states and pursuing alternatives [?], [?]. This work is complementary to VerbalVis: provenance records or reconstructs revisions that have occurred, whereas VerbalVis must determine in real time whether an in-flight action still belongs to the current analytical trajectory.

Taken together, these traditions establish that exploratory analysis can involve evolving goals, revisable explanatory commitments, and progressively refined data focus. They do not establish a universal three-part taxonomy. VerbalVis synthesizes these literatures into three theory-informed, non-exhaustive, and potentially overlapping objects of analytical revision: *Analytical Goal Shift*, *Working-Hypothesis Revision*, and *Analytical Scope Refinement*. The constructs guide the formative study, behavioral coding, and design discussion; they are not runtime classes used to select separate system mechanisms. The unresolved systems question is how a conversational VA environment should remain coherent when any such revision is expressed while speech, analytical tools, and persistent visual state are still active. VerbalVis addresses this gap through response supersession, stale-action invalidation, and replanning from the latest committed dashboard state.

III. FORMATIVE STUDY AND DESIGN REQUIREMENTS

We conducted a lightweight formative design study to understand how analysts express changes in analytical direction during spoken visual analysis and how a conversational visual analytics system should respond when such changes occur. The study was design-oriented rather than confirmatory. It did not aim to discover an exhaustive taxonomy of exploratory behavior, validate a cognitive model, or estimate the preva-

lence of different revision forms. Instead, we used theory-informed concepts as starting points for examining natural expressions of analytical revision, identifying compound and boundary cases, and deriving requirements for coordinating speech, analytical actions, and persistent dashboard state.

A. Study Questions

The study addressed four formative questions:

- FQ1.** How do analysts express changes to their analytical goals, provisional explanations, and data scope during spoken visual analysis?
- FQ2.** When an active analytical response is redirected, what speech, computational work, and dashboard state do users expect the system to preserve, stop, invalidate, or update?
- FQ3.** What boundary cases distinguish analytical revision from recognition correction, conversational repair, ordinary follow-up, evidence seeking, and changes to analytical method or visual representation?
- FQ4.** Which dashboard operations are needed to enact revised analytical directions through conversation?

We define an *analytical revision episode* as a local interaction in which an analyst supersedes, qualifies, or materially modifies an active analytical commitment. The definition concerns the semantic relationship between the latest utterance and the ongoing investigation; it does not require the utterance to occur while the assistant is speaking.

B. Theory-Informed Sensitizing Concepts

The formative inquiry was guided by three sensitizing concepts specified before the sessions. They were synthesized from prior research on sensemaking, exploratory visual analysis, scientific reasoning, information seeking, and visualization tasks. The concepts were therefore not induced solely from the four participants. We used the study to refine their operational boundaries and design implications rather than to validate them as a universal taxonomy.

1) *Analytical Goal Shift*: Sensemaking research characterizes analysis as an iterative process in which people construct and adapt representations in response to task-specific questions [?]. Pirolli and Card similarly describe repeated movement among information gathering, schema construction, hypothesis generation, and the development of analytical products [?]. These general models establish that analytical activity is not necessarily governed by one fixed plan, although they do not themselves define a category called Goal Shift.

More direct support comes from research on evolving information needs and exploratory goals. Bates' berrypicking model describes information seeking as a sequence of evolving queries in which encountered information produces new directions and reformulations [?]. Vakkari shows that search behavior, relevance criteria, and the specificity of information needs change as a research problem develops [?]. In visual analytics, Sacha et al. describe findings and insights as inputs to subsequent exploration and verification loops [?]. Battle

and Heer further synthesize evidence that exploratory visual analysis may begin with broad or underspecified goals that become refined, decomposed, or redirected through interaction with data [?]. Lam et al. distinguish high-level analysis goals from the concrete tasks used to achieve them [?].

Building on these accounts, we use *Analytical Goal Shift* for the stronger case in which an analyst supersedes or materially reorients the primary analytical question or desired knowledge outcome. For example, "I care more about customer ratings than sales" changes what the analyst ultimately seeks to understand. In contrast, changing a chart type, sorting a view, or selecting another analytical procedure does not by itself constitute a goal shift because these changes may preserve the analytical purpose.

2) *Working-Hypothesis Revision*: The strongest and most direct theoretical foundation concerns the revision of provisional explanations. Data-Frame Theory describes sensemaking as a reciprocal relationship between observations and an explanatory frame. As new evidence is encountered, a frame may be elaborated, questioned, compared with alternatives, or replaced [?]. Pontis and Blandford provide empirical support for the development and modification of frames during information-based sensemaking [?].

Scientific-reasoning research makes an additional distinction between search within a hypothesis space and search within an experiment or evidence space [?]. This distinction is important for our coding: changing an explanation is not equivalent to requesting another visualization or test of the same explanation. Research on anomalous evidence further shows that conflicting observations can prompt theory revision, but may also be ignored, reinterpreted, or judged insufficient [?]. Explanatory coherence similarly concerns the evaluation of competing explanations rather than the automatic identification of an objectively correct account [?].

Visual-analytics research also treats hypothesis generation, verification, and falsification as central components of knowledge generation [?]. Systems supporting analytical reasoning externalize relationships among hypotheses, claims, and evidence [?], while empirical work on hypothesis formalization shows that conceptual hypotheses, variables, proxies, and statistical implementations may be iteratively reconsidered [?].

We therefore use *Working-Hypothesis Revision*, rather than *Hypothesis Correction*, for an episode in which the analyst rejects, replaces, qualifies, or materially alters a provisional explanation or causal interpretation. The term *working* emphasizes that the revised explanation remains tentative. For example, "I think the low ratings are caused by delivery delays rather than product quality" revises the active explanation. In contrast, "show me more evidence about delivery delays" requests further evaluation without necessarily changing the working hypothesis.

3) *Analytical Scope Refinement*: A third literature stream describes progressive focusing and the increasing specification of initially broad information needs. Information Foraging Theory explains how people allocate attention to promising information patches and reduce the effective search space as information scent develops [?]. Bates and Vakkari describe the progressive reformulation and specification of queries as

users acquire a clearer understanding of their problem [?], [?]. Battle and Heer similarly describe exploratory visual analysis as moving between open-ended exploration and more focused questions or subtasks [?].

Visualization task research provides a structural basis for distinguishing scope from analytical purpose. Brehmer and Munzner separate the analytical *why*, the interaction *how*, and the data *what* [?]. Lam et al. likewise distinguish broad analysis goals from concrete tasks such as identifying a relevant population or gathering particular evidence [?]. Voyager provides design-oriented support for progressive focus through faceted browsing and partial specification of data and visual properties [?].

These works do not establish Analytical Scope Refinement as a named cognitive category. Rather, they support the underlying phenomenon that analysts progressively specify or alter the data target to which an analytical purpose applies. We operationalize *Analytical Scope Refinement* as a change in constraints over the relevant population, time period, geographic region, variables, categories, granularity, or data subset while the primary analytical objective is generally preserved. Scope narrowing is one subtype; scope broadening, substitution, and changes in granularity are also included.

For example, “continue investigating low ratings, but only for S ao Paulo” preserves the higher-level question while changing its data target. A filter operation is not automatically a scope revision: temporary filtering used to inspect evidence may implement the current plan without changing the intended scope of subsequent reasoning.

4) *Relationships and Boundaries*: The three constructs operate on related but distinguishable objects. Analytical Goal Shift concerns the desired knowledge outcome; Working-Hypothesis Revision concerns an epistemic or explanatory commitment; and Analytical Scope Refinement concerns the data target and constraints. The constructs are not assumed to be mutually exclusive. One utterance may alter several objects simultaneously. For example:

“Forget sales; only examine low-rated orders in S ao Paulo because delivery may be the problem.”

This utterance includes a shift from sales to low ratings, a refinement from the full dataset to low-rated orders in S ao Paulo, and a working hypothesis concerning delivery. Such an episode may therefore receive one primary code and multiple secondary codes.

The constructs are also non-exhaustive. Changes to aggregation, metric, statistical model, visual encoding, chart type, or interaction strategy may constitute a *Method or Representation Change* without changing the analytical goal, working hypothesis, or scope. These cases were retained as residual codes rather than forced into one of the three constructs.

Finally, analytical revision is distinct from barge-in and conversational repair. Barge-in describes when user speech begins relative to system output [?]; analytical revision describes how the semantic content of the new utterance relates to the active investigation. Conversational repair addresses trouble in speaking, hearing, or understanding [?]. A user may therefore interrupt to correct a misrecognized place name

without revising the analysis, or may wait until the assistant finishes before changing the analytical goal.

C. Participants and Apparatus

We recruited four participants (P1–P4) from [RECRUITMENT SOURCE]. All participants had prior experience with data analysis or visualization. They reported [RANGE] years of data-analysis experience and regularly used tools including [ACTUAL TOOLS]. [NUMBER] participants had previously used conversational AI for data-related tasks, and [NUMBER] had experience with speech-based assistants. None had previously used VerbalVis. Participants’ prior familiarity with the Olist dataset was [NONE/LOW/MIXED].

The study used an early prototype of VerbalVis connected to the Olist Brazilian e-commerce dataset. The initial dashboard presented five coordinated views: monthly order volume, category-level sales, state-level distribution, review-score distribution, and delivery-time statistics. The prototype supported spoken requests and a preliminary subset of schema-grounded dashboard operations.

Each session lasted approximately [XX–XX] minutes. Participants received [COMPENSATION]. The study was approved by [ETHICS BODY OR APPROPRIATE ETHICS STATEMENT].

D. Procedure

Each session consisted of five stages.

a) *Introduction and training.*: The researcher introduced the dataset, dashboard views, and speech interface. Participants completed a short practice activity involving filtering, highlighting, and requesting an additional visualization. They were informed that they could redirect the system while it was speaking, but the three sensitizing concepts were not introduced to avoid prompting participants to produce particular revision forms.

b) *Open-ended exploration.*: Participants were asked to investigate [ACTUAL FORMATIVE TASK] and pursue any patterns, questions, or explanations they considered relevant. They could change direction at any time and were not required to interrupt the assistant or produce a prescribed number of revisions. We did not require continuous think-aloud because verbalizing concurrent reasoning could interfere with natural speech interaction. At the end of the activity, participants briefly summarized their findings and analytical path.

c) *Critical-incident replay.*: The researcher replayed selected moments in which the participant redirected, corrected, stopped, or followed up on the assistant. For each incident, the participant was asked:

- what they intended to change;
- which prior analytical state should remain valid;
- whether ongoing speech or analytical work should stop;
- whether already completed dashboard changes should be preserved;
- and how they determined whether the system had followed the latest request.

TABLE I

THEORETICAL FOUNDATIONS OF THE SENSITIZING CONCEPTS USED IN THE FORMATIVE STUDY. PRIOR WORK SUPPORTS THE UNDERLYING OBJECTS OF CHANGE, WHEREAS THEIR ORGANIZATION AS THREE CONSTRUCTS IS OUR SYNTHESIS AND OPERATIONALIZATION FOR CONVERSATIONAL VISUAL ANALYTICS RATHER THAN A PREVIOUSLY ESTABLISHED TAXONOMY.

Construct	Literature foundation	What prior work supports	Operational boundary in this work
Sensemaking, evolving queries, goal evolution, question formulation, and goal-task decomposition [?], [?], [?], [?], [?], [?]	Questions and goals may evolve, become more specific, or open new lines of inquiry as information is encountered.	The primary question or desired knowledge outcome is superseded or materially reoriented. Changes only to visualization, procedure, or interaction strategy are excluded.	Analytical Goal Shift Data-Frame Theory, scientific hypothesis search, anomalous evidence, explanatory coherence, and hypothesis management [?], [?], [?], [?], [?], [?], [?]
Provisional explanations can be generated, evaluated, qualified, compared, rejected, and replaced.	The explanatory proposition changes. Requests for more evidence that preserve the active explanation are excluded.	Working-Hypothesis Revision Information foraging, progressive focusing, query refinement, data-target specification, and faceted exploration [?], [?], [?], [?], [?], [?], [?]	Broad information needs can become focused through the specification of populations, facets, variables, constraints, and levels of detail.
The data target or its constraints change while the higher-level objective is generally preserved. Routine filtering is excluded unless it changes the intended scope of subsequent reasoning.	Analytical Scope Refinement		

d) *Response-lifecycle scenario walkthrough.*: Participants examined short scenarios in which user speech occurred at different points in an analytical response: while the assistant was speaking, after a tool call had been created, during tool execution, and immediately before a dashboard update. Scenarios included analytical redirection, recognition correction, clarification, acknowledgement, evidence seeking, and representation changes. Participants described how they expected the system to respond and which pending actions should remain authorized to modify the dashboard.

e) *Operation elicitation and interview.*: Participants described the dashboard operations they expected to perform through speech and identified actions that would be difficult to express, inspect, or reverse. The interview also covered preferred interruption behavior, visibility of system state, recovery from unintended actions, and the relative importance of immediate cancellation and preservation of completed analytical work.

We recorded system audio, participant speech, transcripts, screen activity, dashboard state, tool events, and researcher notes. For each critical incident, we retained the preceding request, active assistant response, pending analytical actions, participant utterance, and resulting dashboard state.

E. Data Analysis

The analysis followed a hybrid deductive-inductive strategy. Analytical Goal Shift, Working-Hypothesis Revision, and Analytical Scope Refinement were specified before coding as theory-informed sensitizing concepts. We did not assume that all observed behavior would fit these constructs. Open coding was used to identify residual and boundary cases.

We segmented the recordings into candidate episodes whenever a participant spoke in relation to an active analytical

request, assistant explanation, or dashboard operation. Each episode included sufficient context to interpret the new utterance: the preceding user request, active assistant response, visible dashboard state, pending tool actions, and resulting system behavior. Conversation unrelated to the analytical activity was excluded.

The coding scheme included:

- **Analytical Goal Shift;**
- **Working-Hypothesis Revision;**
- **Analytical Scope Refinement;**
- **Method or Representation Change;**
- **Request for Additional Evidence;**
- **Automatic-Speech-Recognition Correction;**
- **Conversational Repair or Clarification;**
- **Ordinary Follow-Up;**
- **Non-Analytical Barge-In;** and
- **Ambiguous or Unclassified.**

An episode could receive one primary code and zero or more secondary codes. The primary code captured the change most necessary to interpret the latest request. Secondary codes captured compound episodes involving multiple objects of change. Interaction timing was coded independently as:

- during assistant speech;
- after speech but before tool completion;
- during tool execution;
- after dashboard commitment; or
- at an ordinary turn boundary.

Two researchers independently coded [ALL EPISODES OR THE ACTUAL PROPORTION]. They first discussed [NUMBER] training episodes to refine the codebook and then coded the remaining material independently. Disagreements were resolved through discussion with reference to the complete interaction context and participants' replay explanations.

Report [THE ACTUAL AGREEMENT PROCEDURE OR RELIABILITY STATISTIC ONLY IF CALCULATED].

Given the formative purpose and small sample, analysis emphasized expressions, boundary cases, user expectations, and design implications rather than frequency comparisons or claims of theoretical saturation.

F. Formative Findings

The analysis yielded [NUMBER] candidate episodes, including [NUMBER] analytical revisions, [NUMBER] repair or clarification episodes, and [NUMBER] other follow-ups or control actions. We report the results as design-oriented themes rather than estimates of the prevalence of revision forms.

1) *Revisions superseded active analytical commitments:* Participants expressed redirection through language such as [INSERT ACTUAL EXPRESSIONS, E.G., ‘ACTUALLY,’ ‘NO, INSTEAD,’ ‘I CARE MORE ABOUT,’ OR ‘ONLY LOOK AT’]. Such utterances did not merely add another operation; they indicated that some aspect of the active analytical response should no longer guide subsequent speech or dashboard updates.

For example, [PARTICIPANT] redirected the analysis from [OLD GOAL] to [NEW GOAL], stating, “[VERBATIM QUOTE].” Another participant retained the question concerning [QUESTION] but revised the explanation from [OLD HYPOTHESIS] to [NEW HYPOTHESIS]. Scope refinements changed [ACTUAL POPULATION/TIME/REGION/VARIABLE] while preserving the broader objective.

These incidents demonstrated the descriptive usefulness of distinguishing the object of change, but they did not establish three exhaustive or mutually exclusive revision types.

2) *Some revision episodes changed multiple objects:* Several utterances combined changes to analytical purpose, explanation, and scope. For instance, [PARTICIPANT] stated, “[ACTUAL COMPOUND QUOTE],” which changed [GOAL], introduced [SCOPE], and proposed [HYPOTHESIS].

These episodes motivated primary and secondary coding rather than forced-choice classification. They also indicated that runtime classification into three types was unnecessary: the system instead needed to identify the latest valid request and prevent response-dependent obsolete work from altering the dashboard.

3) *Stopping speech did not necessarily cancel analytical consequences:* When speech stopped but an earlier operation subsequently modified the dashboard, participants described the behavior as [ACTUAL WORDING]. They expected pending actions associated with the superseded response either to be cancelled or prevented from committing their results.

Participants distinguished pending work from dashboard changes that had already been visibly committed. Completed and still-useful state could remain available, whereas uncommitted results from a superseded response should not appear after redirection. This distinction motivated an explicit separation between the latest committed dashboard state and speculative in-flight work.

4) *Visible dashboard state determined whether the revision appeared successful:* Participants evaluated system understanding through visible filters, highlights, view titles, and created or removed visualizations. A verbal acknowledgement was insufficient when the dashboard continued to reflect the previous request.

Participants requested clearer indications of [ACTUAL REQUESTED STATES, SUCH AS LISTENING, SPEAKING, EXECUTING, OR CANCELLED]. They also emphasized recoverability when a spoken request produced an unintended state change.

5) *Barge-in did not necessarily indicate analytical revision:* The sessions contained interruptions that did not change the analytical goal, working hypothesis, or scope. These included recognition correction, clarification, acknowledgement, requests for additional evidence, and changes to chart type or aggregation.

For example, “[ACTUAL ASR-REPAIR QUOTE]” corrected the system’s recognition, while “[ACTUAL EVIDENCE-REQUEST QUOTE]” requested another test without modifying the active explanation. These cases reinforced the need to separate speech timing from analytical semantics.

6) *Participants requested operations over multiple dashboard targets:* Participant-elicited operations concerned four broad targets:

- **data scope**, including filters, aggregation, and comparison;
- **view specification**, including chart type, encoding, sorting, highlighting, and focus;
- **dashboard organization**, including adding, moving, replacing, and removing views; and
- **analytical assistance**, including explaining a visualization and summarizing the dashboard.

We consolidated these elicited needs with operations identified in established visualization task and interaction frameworks [?], [?] into the conversational dashboard operation inventory in Table ???. The inventory documents a broader design space rather than an exhaustive taxonomy or a list of capabilities fully implemented in VerbalVis. The current prototype implements only the schema-grounded subset used in the subsequent evaluation.

G. Design Requirements

The theoretical framing and formative findings informed five design requirements.

a) *DR1: Separate speech timing from analytical interpretation.*: The system should react promptly when user speech begins but should not assume that every barge-in changes analytical intent. Speech onset may conservatively suspend audio output, while the completed utterance is subsequently interpreted as analytical revision, repair, clarification, evidence seeking, representation change, or ordinary follow-up.

b) *DR2: Coordinate cancellation across speech and analytical execution.*: When a new request supersedes an active response, cancellation should extend beyond audio playback.

Response-dependent tool calls, database operations, visualization specifications, and pending dashboard updates must be cancelled or marked stale so that they cannot later modify the visible state.

c) *DR3: Replan from the latest committed dashboard state.*: The revised response should be grounded in the most recent valid filters, highlights, views, and data scope. Speculative changes associated with a superseded response should not become context for the new plan. The system therefore requires an explicit distinction between committed dashboard state and in-flight analytical work.

d) *DR4: Make listening, execution, cancellation, and commitment visible.*: Users should be able to determine whether the system is listening, speaking, executing an analytical action, cancelling obsolete work, or waiting for input. The interface should expose the state that grounds subsequent conversation and support recovery when an unexpected operation is committed.

e) *DR5: Provide bounded, schema-grounded, and recoverable dashboard operations.*: Conversational interaction should expose a coherent set of operations over data scope, visual specifications, dashboard organization, and analytical assistance. Operations that modify persistent state should use explicit schemas and support removal, replacement, or undo. The broader operation inventory guides future extension, while the present prototype implements the subset required to study full-duplex analytical redirection.

These requirements directly informed the architecture of VerbalVis. DR1 motivated the separation of speech-onset handling from semantic interpretation. DR2 motivated response-bound execution epochs and stale-result rejection. DR3 motivated dashboard versioning and compact state reinjection. DR4 motivated visible listening and execution feedback. DR5 motivated the schema-grounded tool layer described in Section ??.

H. Scope of the Formative Evidence

The formative study provides design-oriented evidence rather than validation of a general theory of analytical intent. Four participants cannot establish theoretical saturation, estimate population-level frequencies, or demonstrate that the three sensitizing concepts cover all exploratory behavior. The constructs may overlap, and revisions to method, metric, operationalization, or visual representation may constitute additional objects of change.

Accordingly, we use the formative study to refine inclusion and exclusion criteria, identify compound and residual cases, and derive requirements for full-duplex conversational visual analytics. The subsequent user study evaluates the interaction and orchestration mechanisms of VerbalVis; it does not test the three constructs as a universal taxonomy.

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. Design Goals

The framework yields three design goals:

- **DG1: Continuous intent revision.** Support revision at any moment, not only at turn boundaries.

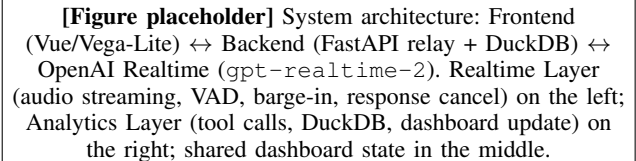


Fig. 2. VerbalVis architecture. The Realtime Layer handles conversational coordination; the Analytics Layer executes tool calls asynchronously and updates the dashboard state, which is reinjected as system context after every tool call.

- **DG2: Rapid replanning after supersession.** Once an interruption is detected during assistant output, the system must prevent stale response work from committing, interpret the redirecting utterance, and re-plan against current dashboard state with minimal added latency.
- **DG3: Dashboard-intent synchrony.** The shared dashboard must always reflect the latest applied operations and must serve as the canonical source of truth for the LLM, so that explanations are grounded in what the user sees rather than in the model’s prior outputs.

B. Architecture Overview

VerbalVis comprises a Vue 3 + Vega-Lite frontend, a FastAPI backend with an in-memory DuckDB analytics layer, and a relay that bridges a browser WebSocket to the OpenAI Realtime WebSocket (Fig. ??). The backend exposes a single /ws endpoint; on connection, it constructs a RealtimeSession that runs two concurrent async loops — one forwarding client audio frames to OpenAI, one demultiplexing OpenAI events to the client and dispatching tool calls.

Realtime layer. gpt-realtime-2 streams 24 kHz PCM audio in both directions. Depending on the experimental condition, the frontend uses local voice gating or push-to-talk, while the Realtime session may also expose server-side speech events such as `input_audio_buffer.speech_started` and `...speech_stopped`. On interruption-relevant events, the backend increments a *turn epoch*, records the in-flight `response_id` as invalidated, and marks tool tasks from the cancelled response as stale. In push-to-talk mode the backend additionally sends `response.cancel`; in server-interruption mode, it relies on the Realtime session’s native interruption behavior and applies the same stale-result discipline locally.

Analytics layer. Tool calls (`response.function_call_arguments.done`) are dispatched to handlers backed by DuckDB queries. DuckDB holds two derived fact tables built from seven Olist CSVs at startup: `fact_order` (one row per delivered order, carrying order-grain attributes including `order_month`, `review_score`, `customer_state`, `delivery_days`, and per-order payment) and `fact_item` (one row per order item, additionally carrying `product_category` and per-item revenue price + freight). The two-table design lets revenue retain the correct semantics under both order-grain and item-grain aggregations; cross-grain filters on

product_category from the order table are rewritten as `order_id IN (SELECT ...FROM fact_item WHERE ...)` subqueries by a shared build_where helper.

VerbalVis employs real-time full-duplex audio interaction for conversational coordination, while analytical operations are executed asynchronously through tool calls that update the dashboard state.

C. Tool Taxonomy

Following the Brehmer–Munzner multi-level typology [?] and Yi et al.’s how-axis [?], we organize VerbalVis operations along three dimensions: *why* (analytical goal), *how* (interaction primitive), and *what* (target of the operation: data, view, layout, insight). Table ?? lists a 15-operation taxonomy for the design space. The current prototype realizes five dashboard operations end-to-end as schema-based tools: `filter_data`, `remove_filter`, `highlight_visual`, `append_visual`, and `delete_visual`. This subset is deliberately narrow: it covers the operations needed for the running example and user-study tasks while keeping tool selection predictable under realtime interruption.

The implemented tools are deliberately minimal. `filter_data` centralizes global dataset narrowing (with a `field=null` convention for clearing, an `append` flag for AND-accumulation, and operator support for `eq`, `neq`, `in`, `gte`, `lte`, `between`); `remove_filter` removes one field constraint without disturbing the rest; `highlight_visual` redirects attention without changing data; `append_visual` creates a new chart at item or order grain based on whether `product_category` appears in the encoding; and `delete_visual` removes views from the shared workspace. This minimality is intentional: most analytical state changes in our scenarios can be expressed by composing these operations, which keeps the model’s tool-selection decision tree shallow and reduces the surface area on which the LLM can hallucinate.

D. Layered Prompt Design

The system prompt is constructed in four labeled sections, in line with the prompt-structure recommendations of the GPT-realtime-2 prompting guide [?]. *Identity* fixes the assistant as a collaborative data analyst, instructs language mirroring, and prescribes the opening behavior (greet, introduce the Olist dataset, walk through the four base views, then invite the user to choose a direction — explicitly without preemptively highlighting any view). *Dashboard Knowledge* provides a bidirectional view-id ↔ Chinese-and-English-label dictionary, the exhaustive field list (forbidding fabricated fields such as “return_rate” or “profit”), and value semantics (e.g., `review_score` 1–2 negative; `customer_state` two-letter codes; revenue in BRL pronounced “reais”). *Tool Usage Rules* specify operator semantics, the cross-grain filter convention, and tool-selection guidelines that explicitly forbid defaulting to the review view for generic prompts such as “what is interesting in the data.” *Realtime Rules* legitimize

interruption as a high-priority signal and instruct the model to abandon prior explanations and re-evaluate against the latest user utterance.

E. Dashboard State and Context Reinjection

The four base views (`view-trend`, `view-review`, `view-map`, `view-category`) and any user-added `workspace-N` charts are maintained as a single backend list. After every tool call, the backend (i) re-queries all views against the current filter set, (ii) recomputes per-view summary statistics (e.g., peak month, low-score ratio, top state, top category), (iii) pushes a `views_update` event to the frontend for re-render, and (iv) injects a compact textual snapshot of the dashboard — highlighted view, active filters, total filtered row count, and per-view summary — back into the LLM as a system message. This *context reinjection* pattern follows the guidance in the GPT-realtime-2 long-context section [?]: the model is never asked to reconstruct dashboard state from prior turns, only to act on what it is given.

F. Supersession Control Protocol

VerbalVis treats barge-in as a control event, not as a semantic classifier. The reusable protocol has five stages. First, the runtime detects possible supersession through either a server speech-start event or a client push-to-talk start during assistant output. Second, the backend increments `turn_epoch`, so newly arriving user input and later tool calls belong to a new response epoch. Third, the system attempts to stop obsolete output through `response.cancel` or `conversation.item.truncate`, depending on the input mode and Realtime event sequence. Fourth, tool calls associated with the superseded response are cancelled when possible and otherwise checked with `is_stale_tool_call` before any result is returned to the model or applied to the dashboard. Fifth, VerbalVis reinjects a compact dashboard snapshot before the next `response.create`, so the revised plan is grounded in the latest valid visual state.

When barge-in is enabled, an interruption is represented by either a server speech-start event or a client push-to-talk start during assistant output. The backend then:

- 1) Increment the per-session `turn_epoch`; record the in-flight `response_id` as invalidated.
- 2) Cancel all in-flight tool tasks bound to the cancelled response.
- 3) Send `response.cancel` or `conversation.item.truncate` when the local input mode requires explicit cancellation or audio truncation.
- 4) Forward `speech_started` to the frontend so the UI can stop playback immediately.

A double-lock discipline (`openai_send_lock` for outbound messages, `tool_state_lock` for state mutation) and a per-response pending-tool counter ensure that (i) tool results from cancelled responses are discarded by an `is_stale_tool_call` check before being forwarded to

TABLE II

VERBALVIS TOOL TAXONOMY. FIVE OPERATIONS ARE IMPLEMENTED END-TO-END AS SCHEMA-BASED TOOLS (MARKED ✓); THE REMAINDER DEFINE THE BROADER DESIGN SPACE. *Why* FOLLOWS BREHMER–MUNZNER; *How* FOLLOWS YI ET AL.; *What* IDENTIFIES THE OPERATION TARGET.

Tool	Why	Why-Detail	How	What	Impl.
filter_data	Discover	Locate	Filter	Data	✓
remove_filter	Discover	Locate	Filter	Data	✓
aggregate_data	Discover	Summarize	Abstract/Elaborate	Data	
compare_data	Discover	Compare	Connect	Data	
change_chart_type	Discover	Explore	Encode	View	
change_encoding	Present	—	Encode	View	
change_sort	Discover	Compare	Reconfigure	View	
highlight_visual	Discover	Locate	Select	View	✓
focus_visual	Discover	Explore	Select+Reconfigure	View	
append_visual	Discover	Explore	Encode	View	✓
move_visual	Present	—	Reconfigure	Layout	
delete_visual	Discover	Identify	Reconfigure	Layout	✓
replace_visual	Discover	Explore	Reconfigure	Layout	
explain_visual	Discover	Identify	Connect	Insight	
summarize_dashboard	Discover	Summarize	Abstract/Elaborate	Insight	

the model, and (ii) when a single model turn issues multiple tool calls, only the final completing call triggers a fresh `response.create`, preventing duplicate or racing responses.

No-barge ablation. We define **VerbalVis-NoBarge** as the controlled ablation of the protocol. It uses the same `gpt-realtime-2` model, voice, prompt, tool schema, dashboard, backend, VAD settings, and context-reinjection logic as **VerbalVis-FullDuplex**. The only intended difference is policy: user speech during assistant output is recorded or queued, but it does not trigger `response.cancel`, `conversation.item.truncate`, epoch invalidation, or stale-tool discard until the current response completes. This condition separates the effect of the realtime model family from the specific effect of the supersession path.

V. CASE STUDY: THREE FORMS OF ANALYTICAL INTENT REVISION

We illustrate the framework with a single session on the Olist Brazilian e-commerce dataset (96,478 delivered orders; 110,196 order items spanning 73 product categories and 27 states; September 2016 – August 2018). The analyst, a domain non-expert, was asked to explore customer satisfaction with no predetermined goal.

A. Initial State

The system greets the user, names the dataset, and walks through the four base views: monthly order trend, review-score distribution, orders by state, top-15 categories by revenue. The system then asks the analyst what they would like to explore. The analyst responds, “Let’s look at the trend.”

B. Type 1 — Goal Shift

System (mid-utterance): “Order volume peaks in November 2017, roughly 4× the average month. Let me decompose that by product category to see whether it’s a Black Friday effect

across categories or concentrated in — ” *Analyst (barge-in):* “I care more about ratings, not sales.”

The barge-in fires `response.cancel`; the redirecting utterance is interpreted; the controller emits `highlight_visual(view-review)` and the assistant continues: “The review distribution is heavily concentrated at 5 stars but with a non-trivial 1- and 2-star tail. Want me to filter to low-rated orders and see what they share?” The intent has shifted from a sales decomposition to a satisfaction investigation; the analyst was spared the remainder of the category breakdown.

C. Type 2 — Hypothesis Correction

System (mid-utterance): “Electronics-adjacent categories such as `computers_accessories` and `electronics` show below-average ratings. This is often attributable to product quality issues with — ” *Analyst (barge-in):* “It’s logistics, not the products.”

The system invokes `append_visual(scatter, delivery_days, review_score)` followed by `filter_data(review_score, lte, 2)`; the resulting scatter shows the low-rated tail clustering at high delivery days. The assistant continues: “You’re right — low-rated orders disproportionately have delivery times above 20 days; the product-quality story doesn’t hold up.” The intent revision substituted a delivery hypothesis for a product-quality hypothesis, and the assistant aborted its prior explanation rather than completing it.

D. Type 3 — Scope Narrowing

System (mid-utterance): “Among states with the worst delivery times, the northeast — Bahia and Ceará — have averages above 25 days, while — ” *Analyst (barge-in):* “Just São Paulo.” `filter_data(customer_state, eq, "SP")` is issued; all four base views recompute against the SP-only subset; the assistant pivots to “Within SP, average delivery is 8.4 days; the low-rated tail is much smaller than the

national figure.” Scope contracted from a national survey to a state-level focus, exploiting the analyst’s prior interest in SP without the analyst having had to articulate that interest at the start of the session.

E. Summary

Each revision corresponds to one of the three framework types; the dashboard, the system narration, and the model’s working hypothesis remained synchronized throughout because every superseded response was followed by a fresh `conversation.item.create` of dashboard context before the next `response.create`. We argue this synchrony is the practical value of treating barge-in as a structural supersession signal rather than merely as audio cancellation.

VI. USER STUDY AND TECHNICAL BENCHMARK

A. Design

The study is designed to test the paper’s central claim rather than general usability: whether a full-duplex supersession pathway makes analytical intent revision in EDA faster, more natural, and more controllable. The primary causal comparison is **not** between a realtime speech model and a cascaded speech pipeline. Instead, we compare two conditions that share the same model, voice, prompt, tool schema, dashboard, backend, VAD configuration, and context-reinjection logic. The only intended difference is whether user speech during assistant output triggers immediate response cancellation, epoch invalidation, and stale-action discard.

We plan a within-subjects, counterbalanced study with $N=12-16$ participants. Participants must have basic data-analysis experience, but need not know the Olist domain in advance. Each participant completes two open-ended exploration tasks on the Olist dataset, one per primary condition; task-condition assignment and condition order are counterbalanced. Each task lasts 12–15 minutes after a short tutorial.

- **Condition A (VerbalVis-FullDuplex):** audio is handled by the `gpt-realtime-2` session; the user can speak during assistant output; barge-in cancels or truncates the current response, increments `turn_epoch`, invalidates stale tool work, and triggers replanning against reinjected dashboard context.
- **Condition B (VerbalVis-NoBarge):** the system uses the same `gpt-realtime-2` session, voice, prompt, VAD settings, dashboard, backend, tool schema, and context-injection logic. However, user speech during assistant output is queued until the current response completes; it does not trigger `response.cancel`, `conversation.item.truncate`, immediate epoch invalidation, or stale-tool discard. This preserves the realtime model family while removing the supersession policy.

Supplementary ecological baseline. We additionally define **VerbalVis-Cascade** as an ecological baseline rather than the main causal comparison. In this condition, user speech is transcribed by ASR, passed to a text LLM using the same dashboard tool schema and Olist backend, and synthesized

back to speech with TTS. The user must wait for the system to finish before the next request is processed. This baseline represents the speech-wrapped, turn-based text-LLM VA regime implicit in systems such as LightVA and DashChat, but it necessarily changes model type, speech pipeline, ASR/TTS latency, audio quality, turn-taking policy, and response timing. We therefore report Cascade results as supplementary context, not as evidence that the barge-in mechanism alone caused any observed advantage.

B. Tasks

Tasks are written to invite open-ended EDA while naturally eliciting the three revision types in our framework. Participants are not instructed to interrupt. The tutorial only explains the interaction regime of each condition: in FullDuplex they may speak whenever they want to redirect the analysis; in NoBarge their speech during assistant output is accepted but processed after the current response finishes. If Cascade is run, participants are told that it is a turn-based speech interface whose next request is processed only after the current response completes.

- **Customer satisfaction task.** “Explore what factors may be associated with low customer review scores. Produce three findings and one follow-up recommendation.”
- **Sales, logistics, and region task.** “Investigate unusual order or revenue patterns and whether they relate to delivery or regional differences. Produce three findings and one follow-up recommendation.”

C. Measurements

Quantitative.

- *Intent revision count* — number of user-initiated direction changes per session, coded from session video, transcript, dashboard events, and the `barge_in.log` timeline.
- *Revision type distribution* — proportion of revisions coded as Goal Shift, Hypothesis Correction, or Scope Narrowing.
- *Exploration breadth* — number of distinct analytical dimensions touched (fields, views, filter expressions) during the session.
- *Insight count* — number of substantive findings reported in the post-task summary, coded by two raters with Cohen’s κ reported.
- *Time-to-revision* — elapsed time between the start of the redirecting utterance and the first system action aligned with the new intent, measured from video and event logs.
- *Wasted output time* — time during which the system continues producing the old analytical direction after the participant has begun redirecting.
- *Queue delay* — in NoBarge, the interval between redirecting speech onset and the moment the queued request begins processing.
- *Time-to-first-audio* (*TTFA*) and *tool-call duration*, recorded automatically by the existing `_response_metrics` instrumentation in `realtime.py`; if Cascade is run, ASR, text-LLM,

and TTS latencies are logged separately and reported as component-level ecological costs.

Qualitative.

- Think-aloud transcripts, coded thematically.
- Semi-structured post-task interviews (10–15 min).
- NASA-TLX (six standard sub-scales).
- Perceived Control scale (3 Likert-7 items, e.g., “I felt able to redirect the analysis whenever I wanted.”).

D. Episode Coding

The unit of analysis is a *revision episode*. Two coders inspect synchronized transcripts, screen recordings, dashboard event logs, and audio timelines. For each candidate episode, coders determine (i) whether it is a valid analytical intent revision rather than a clarification, backchannel, or unrelated speech fragment; (ii) whether it is Goal Shift, Hypothesis Correction, or Scope Narrowing; (iii) whether the system successfully responded to the revised intent; and (iv) the time from the start of the redirecting utterance to the first aligned system action, such as a tool call, dashboard update, or explicit spoken acknowledgment. Disagreements are resolved through discussion after reporting inter-rater agreement.

E. Hypotheses

- **H1:** Intent revision count is higher under VerbalVis-FullDuplex than under VerbalVis-NoBarge.
- **H2:** VerbalVis-FullDuplex reduces time-to-revision, queue delay, and wasted output time relative to VerbalVis-NoBarge.
- **H3:** VerbalVis-FullDuplex increases exploration breadth and insight count relative to VerbalVis-NoBarge.
- **H4:** Perceived control is higher under VerbalVis-FullDuplex, while NASA-TLX is not significantly higher.
- **H5:** The advantage of VerbalVis-FullDuplex is largest for *Hypothesis Correction* and *Scope Narrowing*, because these revisions are most costly when users must wait through an explanation they already intend to reject or constrain.

F. Technical Benchmark

To separate mechanism-level behavior from participant variability, we additionally plan a 250-trial automated benchmark comparing VerbalVis-FullDuplex and VerbalVis-NoBarge. Scripted audio scenarios cover five interruption families: no interruption, early goal shift, late goal shift, hypothesis correction, and scope narrowing. Each scenario is repeated across multiple seeds and dashboard states. The benchmark reports speech-onset detection latency, cancel/truncation latency, stale-tool discard rate, queued-request delay, post-supersession tool correctness, dashboard-context consistency, and component-level latency. The user study tests whether supersession-enabled full-duplex interaction changes analytical behavior and perceived control; the technical benchmark tests whether the mechanism itself reliably cancels stale work, discards stale calls, and preserves dashboard-context consistency.

G. Results (to be reported)

Placeholder. Tables and figures will report per-condition means and within-subject differences for the primary FullDuplex–NoBarge comparison with Wilcoxon signed-rank tests and effect sizes. Cascade, if collected, will be reported in a separate supplementary table to contextualize ecological latency and user experience, not to support the causal barge-in claim. Qualitative themes will be illustrated with representative quotations. The instrumentation already in place yields TTFA, per-turn latency, tool-call duration, stale-call status, and a per-session dashboard context snapshot for every tool invocation, providing both quantitative and trace data.

H. Anticipated Discussion

We expect the analyses to show that (a) the three revision types in our framework appear in the data with distinguishable distributions across participants; (b) the FullDuplex advantage over NoBarge is largest on Hypothesis Correction and Scope Narrowing episodes, where delayed processing forces users to passively witness explanations they already intend to reject or constrain; and (c) Perceived Control correlates positively with intent revision count and negatively with wasted output time. We further expect the automated benchmark to show that the gain is not merely due to the realtime model family, but to the supersession pathway that cancels stale output and preserves dashboard-context consistency.

VII. DISCUSSION

Limitations. The current prototype implements five of the fifteen taxonomy operations as schema-based tools; while these operations cover the running example and planned study tasks, claims about the taxonomy’s completeness require broader deployment. VerbalVis treats all barge-in events with the same downstream processing and does not classify revision type at runtime; the three types in Section ?? are analytical categories, not separate system policies. The planned study size ($N=12-16$) is appropriate for an exploratory within-subjects system study but still limits statistical power for subgroup analysis. VerbalVis-NoBarge is a controlled ablation rather than a naturally deployed product: it preserves the realtime model family while disabling immediate supersession, which is useful for causal attribution but less ecologically familiar. Conversely, VerbalVis-Cascade is ecologically meaningful because it represents a speech-wrapped text-LLM VA regime, but it changes model type, speech pipeline, latency decomposition, audio quality, and turn-taking policy; we therefore treat it only as supplementary context. Finally, the Olist dataset, while rich, is domain-specific; generalization to clinical, financial, or scientific datasets remains future work.

Implications. Treating barge-in as a structural supersession signal expands the conversational VA design space without requiring the system to infer the user’s meaning from timing alone. The interruption event gives the controller permission to protect the analytical state from stale work; the redirecting utterance provides the content of the revised intent. Future systems may also use the *timing* of an interruption (early vs. late in a response) as a weak cue about whether the user

rejects the direction, the evidence, or the conclusion, but such use should remain subordinate to the utterance content and dashboard context.

Generalization beyond voice. The framework is voice-motivated but not voice-bound. Any modality with a true interruption primitive — including streaming-LLM text chat with mid-response edits, or gestural systems that can preempt animations — can be analyzed through the same Intent Evolution lens. We see VerbalVis as a concrete instantiation of a broader design pattern.

Future work. Beyond completing the tool taxonomy and broadening data domains, we plan (i) type-aware interruption handling that differentiates Goal Shift, Hypothesis Correction, and Scope Narrowing at runtime, (ii) a longitudinal study examining whether users’ intent-revision strategies stabilize over repeated sessions, and (iii) an extension to multi-analyst sessions where barge-ins from different participants might conflict or require negotiated replanning.

VIII. CONCLUSION

We have presented VerbalVis, a full-duplex conversational visual analytics system motivated by the observation that user barge-in can serve as a structural supersession signal during analytical work. The Intent Evolution framework identifies three recurring revision types — Goal Shift, Hypothesis Correction, and Scope Narrowing — and motivates a reusable supersession control protocol for conversational EDA. Built on `gpt-realtime-2`, response epochs, schema-based tool calling, and an in-memory DuckDB analytics layer with continuously reinjected dashboard context, VerbalVis demonstrates how full-duplex speech can move intent revision from a between-turn afterthought to a continuously available action. A case study on the Olist e-commerce dataset illustrates all three revision types in a single session; a planned within-subjects user study will make its primary comparison between VerbalVis-FullDuplex and VerbalVis-NoBarge, with VerbalVis-Cascade retained only as a supplementary ecological baseline; and a 250-trial technical benchmark will test the reliability of the supersession mechanism. Together, these evaluations separate the human effect of full-duplex redirection from the mechanism-level question of whether stale work can be canceled or discarded while dashboard context remains consistent.